



# The impact of mobile banking services on saving behavior in West Africa

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## ABSTRACT

This study analyzes the impact of the use of mobile banking services on saving behavior in West Africa. Using the Global Findex Database, 2017 and jointly estimating a multinomial logit model and a probit model, we find that the use of mobile banking services increases the likelihood of formal and informal saving by 2.4% and 0.83% respectively. Women are likelier to have informal savings, but their likelihood of having formal savings increases if they use mobile banking services. Greater education, employment (especially in the public sector), and income increase the likelihood of adopting mobile banking services. These results suggest that the partnership between mobile phone structures and financial institutions should be strengthened.

## 1. Introduction

The role of savings in the financial and economic development of countries has long been established. Saving boosts the autonomy of individuals, households, and enterprises as well as financial institutions and the national economy. Households need to save money not only to reduce their vulnerability to negative shocks such as natural disasters, crop losses, job loss, illness, and disability, but also to make investments.

Theoretically, the study of the savings function has its origins in Keynes, but the saving behavior of households is based on neo-classical life-cycle (Modigliani & Ando, 1957) and permanent income (Friedman, 1957) theories. According to the life-cycle theory, households set a certain level of wealth as a goal for consumption throughout their life. To this motivation, the permanent income hypothesis adds the desire to leave an inheritance. Dupas and Robinson (2013) acknowledge that many people in both developed and developing countries find it difficult to save as much as they would like. However, while households in developed countries have access to many products (certificates of deposit, automatic transfers, etc.) to help them overcome their saving difficulties, households in developing countries tend to resort to much more informal arrangements (Collins, Morduch, Rutherford, & Ruthven, 2009; Rutherford, 2000)—in sub-Saharan Africa, often tontines<sup>2</sup> (Guérin, 2002, 2006) or mobile banks (Lelart & Gnansounou, 1989; Makarimi, 1996). Consequently, the level of formal savings in these countries remains low. Only 21% of adults reported having saved in a formal financial institution (Demirguc-Kunt, Klapper, Singer, Ansar, & Hess, 2017). At the national level, gross domestic savings in the region averaged 19% of the gross domestic product (GDP), compared to 31.6% in North Africa and the Middle East, 27% in South Asia, and almost 43.1% in East Asia and the Pacific (World Bank, 2017). This low level of formal savings in sub-Saharan African countries can be

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<sup>2</sup> A tontine involves no formal contract, but each member pays a fixed amount at regular intervals. The savings collected are allocated to each member in rotation. Members are linked by geographical proximity, gender or age, family or professional relationships, or ethnic group.

explained by low household income levels and frictions in formal financial markets, including transaction costs, lack of trust and regulatory barriers, information and knowledge gaps, social constraints, and behavioral biases (Demirguc-Kunt & Klapper, 2012; Karlan, Ratan, & Zinman, 2014). However, the technological revolution can reduce these constraints (Barr, 2004).

According to many researchers, the use of technology, particularly mobile phones, can improve access to financial products, especially for the poor (Claessens, 2006; Mas & Kumar, 2008). Assadi and Cudi (2011) note that paying through mobile phones avoids the procedures and transaction costs associated with opening a bank account. This technology reduces the transaction costs of banking services from an average of \$4 per transaction to \$0.08 per transaction from a mobile device or the Internet (Grimes, Rogers, & Smith, 2010).

Jack and Suri (2014), Honohan and King (2012), and Mbiti and Weil (2011, 2013) report that mobile phones have evolved from simple communication services to financial tools for both rich and poor households. These tools are less expensive, are easy to use, allow frequent, small transactions, and speed up payment and purchasing—advantages that are likely to improve people's management of resources and consequently increase their savings. Demombynes and Thegeya (2012) show that mobile phones are used as a means of saving by people who are not likely to save through formal channels. Nandhi (2012) claims that innovations with mobile money encourage households to save by minimizing the transaction costs and the risk of informal savings. Moreover, these non-remunerated savings will be increasingly important if they can be easily transferred from a mobile money account to an account at a formal financial institution, which can benefit the saver and the national economy through its role in investment. However, most of the research on the link between financial technologies and the saving behavior of economic agents is limited to the impact of mobile money on savings. The impact of mobile banking services on saving has not been adequately analyzed in sub-Saharan Africa. One study that addressed this issue is that of Ouma, Odongo, and Were (2017), who show that the use of mobile financial services increases the probability of household saving in Kenya. However, this empirical analysis fails to consider the endogeneity of the variable representing use of mobile financial services, which depends on socioeconomic conditions such as income, education, and employment. Moreover, the authors do not distinguish between mobile money and mobile banking services in their analysis.

Thus, the objective of this study is to analyze the impact of mobile banking services on the probability of saving in sub-Saharan African countries, particularly in the West African Economic and Monetary Union (WAEMU). In these countries, mobile banking services are not very widespread in the economic culture, but they raise great hope because of the widespread use of mobile telephone mechanisms in financial transactions. Owing to their ease of use, low transaction costs, and savings of travel and queuing time, these mechanisms have become a driving force for financial inclusion in Africa, as is shown by the success of M-PESA in Kenya. The rate of use of e-money services increased from 0.18% in 2010 to 33.95% in 2018 (BCEAO, 2019)<sup>3</sup>, while the demographic penetration rate of these services increased from 0.31 to 54.53. Between 2017 and 2018 alone, electronic money transactions increased by 14% in volume and 58.30% in value, to 94 million transactions amounting to 10.160 billion F CFA (Francs of the African Financial Community, the local currency) in 2018 (BCEAO, 2019).

Since the launch of the Groupement Interbancaire Monétique (GIM-UEMOA) in 2003, the public authorities of the WAEMU have promoted electronic payment systems to address the lack of nearby financial services and the need for financial inclusion, and to increase the rate of banking. Through the various financial products and services available on mobile phones (mobile banking, mobile payment, SMS banking, rural banking, and branchless banking), the monetary authorities want to encourage savings and channel them towards the most stable and profitable assets. The question then arises whether using mobile banking services changes people's savings behavior.

This paper aims to answer that question. In doing so, it highlights the need for policy makers to strengthen the interaction between mobile money services and formal financial institutions. The study's results will help to improve strategies not only for promoting financial inclusion but also for promoting savings mobilization in the WAEMU countries.

The remainder of this paper is divided into five sections. Section 2 is a review of the literature, and section 3 discusses this study's method. The results are presented in section 4. Section 5 discusses the analysis. The final section concludes the study and discusses the policy implications.

## 2. Literature review

### 2.1. Theoretical analysis

Household saving behavior can be explained by several economic theories, including neoclassical, permanent income, behavioral, and institutional theories. The saving behavior in the neoclassical theory is explained by the life-cycle theory (Modigliani & Ando, 1957; Modigliani & Brumberg, 1954). In this life-cycle theory, savings are justified by the wish to ensure consumption throughout the life cycle, whereas the permanent income hypothesis (Friedman, 1957), retains this motivation but prioritizes the desire to leave an inheritance. All of these theories emphasize household wealth in the tradeoff between consumption and savings. And they all assume that economic agents have perfect information and rationality, whereas behavioral theory explains that economic agents sometimes have difficulty resisting the temptation to spend. As a result, individuals can create their own behavioral incentives and constraints (Shefrin & Thaler, 1988), such as a committed savings account. Mechanisms like payroll deductions make it difficult to choose current consumption at the expense of future consumption (Maital, 1986; Maital & Maital, 1994; Shefrin & Thaler, 1988).

<sup>3</sup> Banque Centrale des Etats de l'Afrique de l'Ouest, is the central bank

Institutional theory also explains saving behavior through incentive mechanisms (Sherraden, 1991). Low- and moderate-income households do not have the same access to institutions that encourage and subsidize asset accumulation, nor do they receive the same incentives from them (Sherraden, 1991; Sherraden, Schreiner, & Beverly, S., 2003). Barr (2004) argues that low-income households often do not have access to bank accounts and face high costs for financial transactions, so it is hard for them to accumulate savings. The author argues that government incentives for financial and technological innovation remove these barriers (Barr, 2004; Beverly & Sherraden, 1999).

## 2.2. Empirical analysis

The relationship between technological innovations and access to financial services has been expounded at both the macro and micro levels. At the macro level, Evans (2018) examines the relationship among the Internet, mobile phones, and financial inclusion in Africa from 2000 to 2016, and finds that the Internet and mobile phones improve people's ability to use basic financial services. The study also shows that macroeconomic factors (such as capital formation, primary schooling, bank credit, broad money, population growth, remittances, agriculture, and interest rates), as well as institutional factors (such as the quality of regulation), are important for financial inclusion in Africa. Using a sample of 52 African countries, Asongu (2012) shows that mobile banking has a negative incidence on traditional financial intermediary dynamics of depth, activity and size, when the conception of a financial system is based only on banks and other financial institution. However, when considering informal financial intermediaries in the analysis, the author shows that mobile banking improves the development of informal financial intermediaries.

At the microeconomic level, a qualitative analysis conducted in Kibera in 2007 by Morawczynski (2009) suggests that the incomes of recipients of mobile remittances in rural areas have increased, as also have household savings. Subsequently, Demombynes and Thegeya (2012) examined mobile savings in Kenya using survey data collected in 2010. They found that the use of mobile savings products embedded in banks remained largely confined to the wealthy; in rural areas, males and married individuals were more likely to save, and most of the population used mobile phones as a depository of funds, especially those who were not likely to save through formal channels. Nandhi (2012) reports that in India, innovations like mobile money encourage households to save by minimizing transaction costs and the risk of informal savings. For Burkina Faso, Ky, Rugemintwari, and Sauviat (2018) analyze the impact of mobile money use on savings in the face of predictable and unpredictable life events. The results show that mobile money use has no correlation with savings for predictable events, but does increase the propensity to save for health emergencies.

These studies reveal that the use of mobile services is mostly limited to mobile money, without being associated with an account in a financial institution. Ouma et al. (2017) examine whether the widespread use of mobile phones to provide financial services is a boon for savings mobilization in Kenya. The results show that the availability and use of mobile phones to provide financial services increases the likelihood of saving at the household level, and also the amount saved. However, the empirical evidence is limited to a single country in East Africa, and, the study design does not take into account the endogeneity of mobile financial service use, nor do the authors distinguish between the impacts of mobile money service and mobile banking service.

## 3. Method

### 3.1. Conceptual framework

The technological revolution, in particular the use of cell phones, computers and the internet, has led to the development of digital finance (Herbst, 2001; Matt, Hess, & Benlian, 2015), such as mobile money, mobile banking, etc. Mobile banking generally refers to new technologies that have enabled access to banking services via mobile devices. The exact definitions, however, differ among researchers. According to Alafeef, Singh, and Ahmad (2012) and Lee and Chung (2009), mobile banking is an m-commerce application that allows customers to use mobile devices to perform activities such as checking their bank account status, transferring money, making payments, or selling shares. Others (Akturan & Tezcan, 2012; Masrek, Omar, Uzir, & Khairuddin, 2012; Shih, Hung, & Lin, 2010) define mobile banking generally as an innovative communication channel through which the customer interacts with a bank using a mobile device. In a narrower sense, mobile banking can be defined as the use of cell phones to provide formal financial services. In a broader sense, it includes payment services commonly referred to as m-payments, internet banking, and other miscellaneous services that are made possible through cell phones (Lin, 2011). It should be noted that the concept of mobile banking is different from mobile money. Mobile money is a service offered by an electronic money issuer to its customers, allowing them to make financial transactions using their cell phone. In West Africa, mobile money service includes communication recharging, money transfer and deposit, bill payment, and internet package purchase. In addition to these services, mobile banking offers internet banking and other miscellaneous bank services (savings, loans, insurance) that are made possible through mobile devices. Mobile money users can deposit funds for free, but are charged for withdrawals and transfers.

In West Africa, the central bank sets limits<sup>4</sup> for such transactions of mobile money (both deposits and withdrawals) and monitors the related activities, and all transactions require a national identity card.

Fig. 1 shows how mobile banking can affect the financial and saving behavior of its users. Mobile banking provides benefits such as

<sup>4</sup> Article 31 of BCEAO Instruction no. 008-05-2015, dated May 21, 2015, has set the ceiling amount to be held by one person per day, other than distributors and acceptors, at 2 million FCFA. When a person holds several accounts, the ceiling applies to the total balance. The cumulative supply of electronic money is limited to 10 million FCFA per month.

reduced transaction costs, accessibility, convenience, reliability, and security in financial transactions. These benefits can change the financial behavior of economic agents by enabling them to manage their resources rationally and improve their savings levels. Making deposits, withdrawals, payments, purchases, and transfers using a mobile device reduces travel costs and waiting times at counters; it also saves time for business activities. Moreover, accessibility anywhere and at any time makes it possible to track account movements, adjust spending according to available resources, and set a savings target.

In addition to providing convenience, reliability, and security, mobile banking makes financial transactions traceable and transparent, increasing trust between economic agents and banks—an important factor that can change financial behavior, especially saving behavior.

### 3.2. The empirical model

Most of the studies on access to financial services use either probit or logit models (Allen, Demircuc-Kunt, Klapper, & Martinez Peria, 2012; Soumaré, Tchana Tchana, & Kengne, 2016) with binary dependent variables. However, savings is not simply a binary choice; individuals can choose from an array of mechanisms, both formal (an account with a bank, microfinance institution, or rural cooperative) and informal (a community savings group, cash under the mattress, jewelry, or livestock) (Dupas & Robinson, 2013; Karlan et al., 2014; Klapper & Singer, 2014). Following Murendo and Mutsonziwa (2017), we have used a multinomial logit model to assess the effect of mobile banking on formal and informal saving.

Let  $Y_i$  be a variable that expresses saving.

$$Y_i = \begin{cases} 0 & \text{if the individual had not saved} \\ 1 & \text{if the individual had saved in a formal financial institution} \\ 2 & \text{if the individual had saved in an informal club} \\ 3 & \text{if the individual had saved in both formal and informal institutions} \end{cases} \quad (1)$$

The estimation model is as follows:

$$Y_{ij} = \alpha_0 + \alpha_{1j}T_i + \alpha_{2j}X_i + \epsilon_{ij} \quad (2)$$

where  $Y_{i,j}$  is the savings choice  $j$  of individual  $i$ ,  $X$  is a vector of the socioeconomic characteristics of individual  $i$ ,  $T$  is the use of mobile banking service, and  $\alpha_1$  the impact of the use of mobile banking service on the probability of saving. The matrix of control variables  $X$  is composed of age, education, income level, occupation status, and line of business, while  $\epsilon$  is the error term. The use of mobile banking service,  $T$ , is a binary variable defined as follows:

$$T_i = \begin{cases} 1 & \text{if the individual used a mobile banking service} \\ 0 & \text{otherwise} \end{cases} \quad (3)$$

### 3.3. Estimation method

#### 3.3.1. The problem of endogeneity

In analyzing the impact of mobile banking use on saving choice, we must take into account two possible sources of endogeneity. First, the decision to use mobile services may be correlated both with observable characteristics, such as income level, education level, and employment status, and with unobservable characteristics. This results in bias to the extent that individuals have heterogeneous characteristics (Seng, 2017). Moreover, the choice to adopt mobile banking services may be influenced by household saving behavior (Ky et al., 2018)—a problem of double causality. Therefore, the equation for use of mobile banking can be written as

$$T_i = \beta_0 + \beta_1 Z_i + \delta X_i + u_i \quad (4)$$

where  $Z_i$  represents the matrix of variables other than  $X$ ;  $u_i$  is the error term. Assuming that the error terms follow a normal distribution, we chose a probit model.

#### 3.3.2. Estimation procedure

In the presence of endogeneity, classical methods can bias the estimated coefficients. To correct for this potential bias, a generalized structural equation model estimation (GSEM) is appropriate (Bartus, 2017). This method allows the estimation of a multinomial logit model with an endogenous treatment variable. It is a full information maximum likelihood (FIML) method of estimation that controls endogeneity issues related to the estimation of mixed models. In this case, Eqs. (2) and (4) are jointly estimated. This method makes it possible to obtain both the determinants of the use of mobile banking service and the effect of that service on saving.

### 3.4. Data and variable definitions

#### 3.4.1. Data

The data used in this study are from the World Bank's Global Findex, 2017 database. They were collected from 148 countries, with 1000 people surveyed in each country. The objective of these surveys was to assess the level of financial inclusion in terms of both access to and use of financial services. The sample was nationally representative, the target population consisted of the total civilian

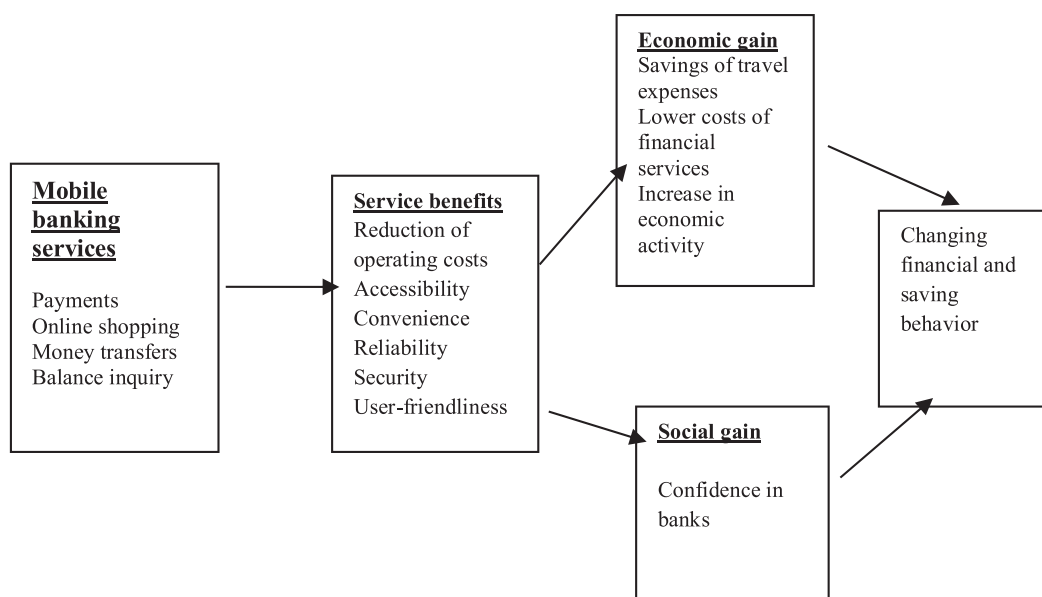


Fig. 1. Relationship between mobile banking services and saving behavior.<sup>11</sup>

non institutional population aged 15 years and above, and the unit of observation was the individual (Demirguc-Kunt & Klapper, 2012). In our analysis, we consider seven West African countries—Benin, Burkina Faso, Ivory Coast, Mali, Niger, Senegal, and Togo—that are part of a monetary union. This region has a low rate of “bancarization” at 17.7% (BCEAO, 2017), but the countries have the same banking networks and almost similar banking services in addition to a common monetary policy,<sup>5</sup> making them good candidates for the development of electronic banking. The sample used for the estimation comprised 6074 observations.

### 3.4.2. Variables

Variables are described in Table 1. The dependent variable is an individual’s savings during the past 12 months, for the purpose of future spending on education, marriage, and major purchases, or emergencies that may occur when the individual is no longer working. Following Klapper and Singer (2014), we specify the variable in four levels: none; formal savings; informal savings; both formal and informal savings.

### 3.4.3. The treatment variable

The treatment variable is using a mobile device for transactions (payments, purchases, money transfers) from an account at a formal financial institution. The choice of this indicator makes it possible to assess, on the one hand, the interaction between mobile telephone services and the services of formal financial institutions and, on the other hand, the financial behavior—particularly the saving behavior—of individuals using these services. This choice also responds to the monetary authorities’ concern about lack of proximity to financial services in the WAEMU countries. The other exogenous variables, draw on the literature on financial inclusion (Allen et al., 2012; Aterido, Beck, & Iacovone, 2013; Demirguc-Kunt, Klapper, & Singer, 2013; Soumaré et al., 2016).

## 4. Results

### 4.1. Descriptive statistics

Regarding the sample’s demographic characteristics, 41% of the respondents were female and 59% were male (Table 2). The average age was 32 years. The highest level of education attained was primary education for 61.2% of the respondents, secondary education for 36.2%, and higher education for only 2.6%. Regarding employment status, 67.6% of the respondents were employed, and 8.4% were self-employed.

Among the five income brackets, 15.7% of respondents belonged to poorest, 16.9% to second, 18.7% to middle, 21.1% to fourth, and 27.7% to richest. For 5.2%, most of this income came from the public sector; for 25.4%, from the agricultural sector. The statistics revealed that 64.3% of respondents had a national ID, and 72.8% had a mobile phone.

Source: World bank (2017) and author’s calculations.

In the last 12 months, 27.8% of the respondents had saved: 13.5% for business reasons, 5.2% for old age, and 5.6% for both old age

<sup>5</sup> Banks in WAEMU countries are governed by a set of texts and regulations that apply uniformly.

**Table 1**  
Description of variables.

| Variables                  | Definition                                                                                                                              |
|----------------------------|-----------------------------------------------------------------------------------------------------------------------------------------|
| Dependent variable         |                                                                                                                                         |
| Formal saving              | 1 if an individual has saved in a formal financial institution during the past 12 months and 0 if not.                                  |
| Informal saving            | 1 if an individual has saved in an informal financial institution during the past 12 months and 0 if not.                               |
| Formal and informal saving | 1 if an individual has saved in both formal and informal institutions during the past 12 months and 0 if not.                           |
| Saving for business        | 1 if an individual has saved during the past 12 months for business and 0 if not.                                                       |
| Saving for old age         | 1 if an individual has saved during the past 12 months for old age and 0 if not.                                                        |
| Treatment variable         |                                                                                                                                         |
| Mobile banking             | 1 if the individual used mobile banking service and 0 if not.                                                                           |
| Control variables          |                                                                                                                                         |
| Sex                        | 1 if the individual is female and 0 if he is male.                                                                                      |
| Education                  | 1 if the individual has attained a certain level of education and 0 if not, for each of three levels: primary, secondary, and tertiary. |
| Income level               | Five income brackets corresponding to income quintiles, namely poorest 20%, second 20%, middle 20%, fourth 20% and richest 20%.         |
| Public sector              | 1 if the individual gets most of his/her income from the public sector and 0 if not.                                                    |
| Agricultural sector        | 1 if the individual gets most of his/her income from the agricultural sector and 0 if not.                                              |
| Self-employed              | 1 if the individual is self-employed and 0 if not.                                                                                      |
| Employed                   | 1 if the individual works and 0 if not.                                                                                                 |
| Age                        | in completed years.                                                                                                                     |
| Mobile phone               | 1 if the individual has a mobile phone and 0 if not.                                                                                    |
| National ID                | 1 if the individual has a national ID and 0 otherwise.                                                                                  |
| Mobile money               | 1 if the individual used mobile money and 0 if not.                                                                                     |

**Table 2**  
Descriptive statistics.

| Variable                             | Mean   | User   | Non user | Mean test |
|--------------------------------------|--------|--------|----------|-----------|
| Mobile banking                       | 0.047  |        |          |           |
| Saving                               | 0.278  | 0.744  | 0.254    | 0.490**** |
| Saving for business                  | 0.135  | 0.237  | 0.130    | 0.107***  |
| Saving for old age                   | 0.052  | 0.100  | 0.050    | 0.049***  |
| Saving for both old age and business | 0.056  | 0.244  | 0.047    | 0.197***  |
| Formal saving                        | 0.052  | 0.317  | 0.039    | 0.278***  |
| Informal saving                      | 0.195  | 0.175  | 0.196    | -0.020    |
| Formal and informal saving           | 0.030  | 0.251  | 0.019    | 0.232***  |
| Age                                  | 32.623 | 32.541 | 32.627   | -0.086    |
| Sex                                  | 0.410  | 0.227  | 0.418    | -0.191*** |
| Employed                             | 0.676  | 0.863  | 0.667    | 0.196***  |
| Primary ed.                          | 0.612  | 0.246  | 0.628    | -0.382*** |
| Secondary ed.                        | 0.362  | 0.606  | 0.352    | 0.254***  |
| Tertiary ed.                         | 0.026  | 0.147  | 0.019    | 0.128***  |
| Richest 20%                          | 0.277  | 0.520  | 0.264    | 0.256***  |
| Fourth 20%                           | 0.211  | 0.185  | 0.212    | -0.027    |
| Middle 20%                           | 0.187  | 0.127  | 0.189    | -0.062*** |
| Second 20%                           | 0.169  | 0.109  | 0.172    | -0.063*** |
| Poorest 20%                          | 0.157  | 0.058  | 0.162    | -0.104    |
| Public sector                        | 0.052  | 0.246  | 0.042    | 0.204***  |
| Agricultural sector                  | 0.254  | 0.284  | 0.253    | 0.031     |
| Self-employed                        | 0.084  | 0.127  | 0.082    | 0.045***  |
| National ID                          | 0.643  | 0.935  | 0.628    | 0.307***  |
| Mobile phone                         | 0.728  | 0.941  | 0.718    | 0.223***  |
| Mobile money                         | 0.368  | 1.000  | 0.374    | 0.344***  |

\*\*\* p < 0.01.

\*\* p < 0.05.

and business reasons. Among savers, 5.24% of the whole sample had saved in formal institutions, 19.54% in informal clubs, and 3.03% in both formal and informal institutions.

Mobile banking use was low, at 4.7%, while 36.89% used mobile money. However, 100% of mobile banking users also used mobile money, as did 37.4% of non users of mobile banking. Mobile banking users were significantly likelier to have all forms of savings except informal savings.

<sup>1</sup> source: author

## 4.2. Econometric results

The results of the estimation of the generalized structural equation model are presented in Table 3. It is a joint estimation of both the multinomial logit and probit models by the full information maximum likelihood method.

The results indicate that the use of mobile banking increases the likelihood of saving through both formal and informal mechanisms by 3.14%, the likelihood of saving through formal mechanisms by 2.40%, and the likelihood of saving through informal mechanisms by only 0.83%.

Some control variables affect the likelihood of saving. Secondary and tertiary education and high income increase the likelihood of formal saving. A job, whether in the public or agricultural sector or in self-employment, increases the probability of both formal and informal saving, as does age up to a given threshold. Being female significantly increases informal savings and significantly decreases formal savings. But if women use mobile banking services, they increase their likelihood of formal saving; the coefficient of the interaction variable is positive and significant. In fact, an increase of 1 point in women's use of mobile banking increases their likelihood of saving by 0.94 points. Respondents are likelier to save for business (0.78) than for old age (0.71) (Table 4).

In addition, given the interest in mobile money in West African countries, we have compared its effect on different forms of savings with that of mobile banking (see Table 5).

Both mobile banking and mobile money increase the probability of both formal and informal saving, but mobile banking increases the likelihood of formal saving more (1.54) than mobile money does (0.90). On the other hand, the use of mobile money increases the likelihood of informal saving more (0.59) than mobile banking does (0.47). The simultaneous use of mobile money and mobile banking increases the likelihood of formal savings. The adoption of mobile banking is positively correlated with education, high income, working, public service employment, agricultural sector employment, self-employment, ID number ownership, and mobile phone ownership.

## 5. Discussion

### 5.1. Determinants of mobile banking services

Socioeconomic factors such as working, education, high income, and the status of worker, whatever the sector of activity, increase the probability of adopting mobile banking services. But employment in the public sector has more effect than employment in other sectors—not surprisingly, as in the countries we analyze, public-sector employment offers greater stability and requires more education, which in turn makes it easier for the employee to understand and adopt innovations in financial services (Delavande, Rohwedder, & Willis, 2008; Hibbert, Lawrence, & Prakash, 2013; Jappelli & Padula, 2013). This result is consistent with the work of Fall, Ky, and Birba (2015) on the adoption of m-banking in Senegal. Similarly, Mbiti and Weil (2011) found that Kenyans with higher socioeconomic status were more likely to adopt financial innovations. Obviously, to use mobile banking one must have a mobile device

**Table 3**  
Mobile banking and saving.

|                     | Saving determinants |           |                 |           |                            |           | Use of mobile banking |           |
|---------------------|---------------------|-----------|-----------------|-----------|----------------------------|-----------|-----------------------|-----------|
|                     | Formal saving       |           | Informal saving |           | Formal and informal saving |           | coef                  | Std. Err. |
|                     | Coef                | Std. Err. | coef            | Std. Err. | coef                       | Std. Err. |                       |           |
| Mobile banking      | 2.244***            | 0.205     | 0.700***        | 0.223     | 3.057***                   | 0.244     |                       |           |
| Age                 | 0.140***            | 0.028     | 0.061***        | 0.013     | 0.104***                   | 0.037     | 0.031**               | 0.014     |
| Age <sup>2</sup>    | -0.001***           | 0.000     | -0.001***       | 0.000     | -0.001**                   | 0.000     | -0.000**              | 0.000     |
| Employed            | 0.788***            | 0.180     | 0.491***        | 0.078     | 0.966***                   | 0.237     | 0.371***              | 0.085     |
| Sex                 | -0.513***           | 0.182     | 0.619***        | 0.071     | 0.755***                   | 0.202     | -0.186***             | 0.073     |
| Sex*Mobile banking  | 0.946*              | 0.501     | 0.613           | 0.455     | 0.490                      | 0.487     |                       |           |
| Secondary ed.       | 0.994***            | 0.153     | 0.142*          | 0.076     | 0.535***                   | 0.191     | 0.476***              | 0.073     |
| Tertiary ed.        | 1.455***            | 0.270     | 0.006           | 0.250     | 0.736**                    | 0.366     | 1.011***              | 0.129     |
| Richest 20%         | 0.917***            | 0.207     | 0.023           | 0.102     | 1.356***                   | 0.310     | 0.182*                | 0.096     |
| Fourth 20%          | 0.332               | 0.232     | 0.108           | 0.103     | 0.895**                    | 0.334     | 0.014                 | 0.107     |
| Second 20%          | -0.246              | 0.282     | -0.183          | 0.113     | 0.930***                   | 0.344     | 0.035                 | 0.118     |
| Poorest 20%         | -0.574*             | 0.332     | -0.281**        | 0.117     | 0.199                      | 0.412     | -0.150                | 0.137     |
| Public sector       | 1.554***            | 0.187     | 0.815***        | 0.158     | 1.658***                   | 0.231     | 0.585***              | 0.092     |
| Agricultural sector | 0.515***            | 0.153     | 0.507***        | 0.078     | 0.902***                   | 0.182     | 0.145**               | 0.074     |
| Self-employed       | 1.109***            | 0.190     | 0.830***        | 0.112     | 0.656**                    | 0.287     | 0.192*                | 0.102     |
| Mobile phone        |                     |           |                 |           |                            |           | 0.337***              | 0.110     |
| National ID         |                     |           |                 |           |                            |           | 0.640***              | 0.101     |
| cons                | -7.608***           | 0.580     | -3.193***       | 0.245     | -8.147***                  | 0.764     | -3.493***             | 0.276     |
| Log likelihood      | -5251.669           |           |                 |           |                            |           |                       |           |
| N                   | 6074                |           |                 |           |                            |           |                       |           |

Source: author's calculations.

\*\*\* p < 0.01.

\*\* p < 0.05.

\* p < 0.1.

**Table 4**  
Mobile banking and saving choices.

|                     | Saving determinants |           |                    |           |                                      |           | Use of Mobile banking |           |
|---------------------|---------------------|-----------|--------------------|-----------|--------------------------------------|-----------|-----------------------|-----------|
|                     | Saving for business |           | Saving for old age |           | Saving for both business and old age |           | coef                  | Std. Err. |
|                     | Coef                | Std. Err. | coef               | Std. Err. | coef                                 | Std. Err. |                       |           |
| Mobile banking      | 0.786***            | 0.169     | 0.712***           | 0.229     | 1.690***                             | 0.182     |                       |           |
| Age                 | 0.065***            | 0.016     | 0.031              | 0.019     | 0.074***                             | 0.022     | 0.031**               | 0.014     |
| Age <sup>2</sup>    | -0.001***           | 0.000     | -0.000             | 0.000     | -0.001***                            | 0.000     | -0.000**              | 0.000     |
| Employed            | 1.089***            | 0.111     | 0.309**            | 0.136     | 0.635***                             | 0.156     | 0.371***              | 0.085     |
| Sex                 | -0.131              | 0.085     | -0.151             | 0.129     | -0.200                               | 0.132     | -0.186***             | 0.073     |
| Secondary ed.       | 0.101               | 0.090     | 0.502***           | 0.135     | -0.021                               | 0.137     | 0.476***              | 0.073     |
| Tertiary ed.        | 0.369               | 0.233     | 0.350              | 0.324     | 0.039                                | 0.306     | 1.011***              | 0.129     |
| Richest 20%         | 0.393***            | 0.122     | 0.900***           | 0.196     | 0.649***                             | 0.180     | 0.182*                | 0.096     |
| Fourth 20%          | 0.212*              | 0.129     | 0.499**            | 0.212     | 0.323*                               | 0.194     | 0.014                 | 0.107     |
| Second 20%          | 0.191               | 0.135     | 0.413*             | 0.227     | 0.128                                | 0.210     | 0.035                 | 0.118     |
| Poorest 20%         | 0.100               | 0.142     | -0.160             | 0.265     | -0.409                               | 0.254     | -0.150                | 0.137     |
| Public sector       | 0.614***            | 0.168     | 1.097***           | 0.197     | 1.467***                             | 0.186     | 0.585***              | 0.092     |
| Agricultural sector | 1.102***            | 0.088     | 0.309**            | 0.143     | 1.260***                             | 0.132     | 0.145**               | 0.074     |
| Self-employed       | 1.229***            | 0.123     | 0.488**            | 0.206     | 1.413***                             | 0.182     | 0.192*                | 0.102     |
| Mobile phone        |                     |           |                    |           |                                      |           | 0.337***              | 0.110     |
| National ID         |                     |           |                    |           |                                      |           | 0.640***              | 0.101     |
| cons                | -4.382***           | 0.308     | -4.571***          | 0.413     | -5.566***                            | 0.459     | -3.493***             | 0.276     |
| Log likelihood      | -5336.762           |           |                    |           |                                      |           |                       |           |
| N                   | 6074                |           |                    |           |                                      |           |                       |           |

Source: author's calculations.

\*\*\* p < 0.01.

\*\* p < 0.05.

\* p < 0.1.

**Table 5**  
Mobile banking vs. mobile money: effects on saving.

|                              | Saving determinants |           |                 |           |                            |           | Use of Mobile banking |           |
|------------------------------|---------------------|-----------|-----------------|-----------|----------------------------|-----------|-----------------------|-----------|
|                              | Formal saving       |           | Informal saving |           | Formal and informal saving |           | coef                  | Std. Err. |
|                              | Coef                | Std. Err. | Coef            | Std. Err. | coef                       | Std. Err. |                       |           |
| Mobile banking               | 1.546***            | 0.271     | 0.478*          | 0.264     | 2.324***                   | 0.291     |                       |           |
| Mobile money                 | 0.908***            | 0.151     | 0.594***        | 0.073     | 1.237***                   | 0.210     |                       |           |
| Mobile banking* Mobile money | 0.784**             | 0.345     | 0.084           | 0.371     | 0.521                      | 0.356     |                       |           |
| Age                          | 0.141***            | 0.028     | 0.057***        | 0.013     | 0.102***                   | 0.037     | .031**                | 0.014     |
| Age <sup>2</sup>             | -0.001***           | 0.000     | -0.001***       | 0.000     | -0.001**                   | 0.000     | -0.000**              | 0.000     |
| Employed                     | 0.714***            | 0.182     | 0.444***        | 0.079     | 0.873***                   | 0.239     | 0.371***              | 0.085     |
| Sex                          | -0.357**            | 0.164     | 0.661***        | 0.071     | 0.819***                   | 0.177     | -0.186***             | 0.073     |
| Secondary ed.                | 0.834***            | 0.156     | 0.025           | 0.078     | 0.315                      | 0.195     | 0.476***              | 0.073     |
| Tertiary ed.                 | 1.267***            | 0.272     | -0.137          | 0.251     | 0.496                      | 0.368     | 1.011***              | 0.129     |
| Richest 20%                  | 0.895***            | 0.208     | -0.003          | 0.103     | 1.314***                   | 0.312     | 0.182*                | 0.096     |
| Fourth 20%                   | 0.314               | 0.233     | 0.093           | 0.104     | 0.858**                    | 0.336     | 0.014                 | 0.107     |
| Second 20%                   | -0.264              | 0.284     | -0.184          | 0.113     | 0.919***                   | 0.346     | 0.035                 | 0.118     |
| Poorest 20%                  | -0.521              | 0.334     | -0.241**        | 0.117     | 0.280                      | 0.414     | -0.150                | 0.137     |
| Public sector                | 1.478***            | 0.187     | 0.735***        | 0.160     | 1.547***                   | 0.230     | 0.585***              | 0.092     |
| Agricultural sector          | 0.457***            | 0.154     | 0.477***        | 0.078     | 0.829***                   | 0.183     | 0.145**               | 0.074     |
| Self-employed                | 1.041***            | 0.191     | 0.761***        | 0.114     | 0.550*                     | 0.287     | 0.192*                | 0.102     |
| Mobile phone                 |                     |           |                 |           |                            |           | 0.337***              | 0.110     |
| National ID                  |                     |           |                 |           |                            |           | 0.640***              | 0.101     |
| cons                         | -7.938***           | 0.592     | -3.278***       | 0.248     | -8.533***                  | 0.775     | -3.493***             | 0.276     |
| Log likelihood               | -5192.769           |           |                 |           |                            |           |                       |           |
| N                            | 6074                |           |                 |           |                            |           |                       |           |

Source: author's calculations.

\*\*\* p < 0.01.

\*\* p < 0.05.

\* p < 0.1.

such as a cell phone, as well as an identification number. Claessens (2006) shows in fact that mobile phones can improve access to financial products.

## 5.2. Effects of mobile banking services on saving

The results indicate that the use of mobile banking increases the likelihood of formal and informal savings. By definition, of course, people who have access to mobile banking have the possibility of formal savings. And in West Africa, there are strong incentives: a savings account carries interest at a minimum rate of 3.5% (BCEAO, 2018), while the cost of bank credit to carry out a long-term project is high (at least 10% according to BCEAO, 2018).

In addition, mobile banking services reduce travel costs. In West African countries, most banks are located in densely populated large cities that are administrative and commercial centers. Rural areas, which are essentially agricultural and lack adequate infrastructure, are not of interest to these banks. The distance from a bank may discourage households from adopting formal financial products, in particular savings products. Mobile banking offers the possibility of conducting financial transactions anytime and anywhere, eliminating the need to visit a bank branch or long-queued bill payment agencies. The frequency and convenience with which financial transactions can be carried out using a mobile phone may induce people to reallocate their resources between consumption and saving—especially since money stored in mobile accounts earns no interest. In the WAEMU countries, with the development of mobile money, some people receive most of their payments through their mobile accounts; indeed, according to central bank statistics such payments account for 33.95% of the usage of those accounts. Because some banks offer a link between mobile money accounts and bank accounts, this income can be easily and permanently transferred, avoiding the risk of losing money through informal mechanisms like tontines. And the ability to track bank account transactions from the mobile phone allows people to adjust their expenses to their income, and thus to modify their savings. Our results are in line with institutional theory, which shows the important role of technology in access to formal financial services, including savings. These results are also consistent with those found by Ouma et al. (2017) in Kenya, which indicate that the use of mobile financial services increases the probability and the amount of savings. These results imply that the development of these services should be strengthened by further encouraging the partnership between mobile phones and financial institutions.

Several factors may explain the positive effect of mobile banking on informal savings. In West African countries, many economic agents resort to informal ways of saving, such as informal tontines (Guérin, 2006; Lelart & Gnansounou, 1989); purchase and storage of livestock or gold jewelry; and cash at home “under a mattress.” These informal savings are held to meet unpredictable needs, such as social events (baptisms, marriages, deaths, etc.) or health problems. Similarly, as Morawczynski (2009) explain, individuals use their mobile money accounts for emergencies, while for long-term savings they prefer tools that limit access to liquidity, such as bank accounts. And mobile banking allows transfers from a bank account to the mobile money account; indeed, statistics show that mobile banking users also use mobile money.

Saving for business is commoner than saving for old age. In West Africa it is difficult to get formal financing to undertake a business, owing to the lack of information-sharing structures (Asongu, 2017); instead, aspiring business people must save (Machokoto & Areneke, 2020).

The older the individual, the more likely it is that he or she works, and in turn has an account in a financial institution, either to receive a salary or to accumulate savings. However, our results indicate that beyond a threshold level the probability to save decreases. This result is similar to Ouma et al. (2017) finding and is consistent with the life-cycle hypothesis, which indicates that when an individual is in the labor force, he or she accumulates savings, whereas after retirement he or she spends those savings to ensure consumption. Education increases one's chances of professional integration into the labor market and thus one's demand for financial services, including savings mechanisms. These results imply that education should be further encouraged. Employment in any sector increases the likelihood of saving in formal and informal institutions—not surprisingly, since neoclassical theory assigns an important role to income in saving decisions.

The negative effect of gender on formal savings, and its positive effect on informal savings, is an interesting result. Women in West African countries are marginalized from the formal financial market by socioeconomic factors such as marriage, domestic work, motherhood, and elder care that prevent them from pursuing education and participating effectively in the formal labor market. They are disproportionately represented among the 76% of the workforce that is engaged in low-paying informal activities (Brilleau, Roubaud, & Torelli, 2004). In addition, access to credit in West Africa for the most part requires a guarantee (salary, land, or livestock). Even if legislation allows women to hold land, prejudice and customary law mean that they are rarely full owners (Alinon, 2008). But if women use mobile banking, their likelihood of formal saving increases. The ease of the technology makes it possible even for low-income people to accumulate formal savings. As Barr (2004) points out, financial innovations provide savings opportunities for households, especially the poor. And Fall and Birba (2019) find that once women adopt mobile banking, they are more likely to continue to use it. Our result implies that banking institutions can improve savings mobilization by developing micro-savings products adapted to the working and income conditions of women and low-income people. This requires reduction traditional savings conditions, such as minimum balances, and fees for transferring money from the mobile money account to the bank account. In addition, to increase the mobilization of savings, it would be interesting to apply competitive interest rates and easy access to credit.

Our finding that the use of mobile money increases the likelihood of informal saving more than mobile banking does is explained by the costs of opening bank accounts and the distance between low-income households and banks. Mobile money has become an informal savings mechanism to replace savings in the form of livestock, jewelry, and tontines, as it ensures the security of money through the protection of a security code. Ky et al. (2018) found that in Burkina Faso the use of mobile money increased the propensity to save for health emergencies, especially among disadvantaged groups such as rural people, women, less educated people, and people

with irregular income. Our results also show that the simultaneous use of mobile money and mobile banking increases the likelihood of formal savings. Indeed, an economic agent who has access to mobile banking will prefer formal savings that are remunerated, but will use mobile money for other services, such as transfers and payments.

## 6. Conclusion and policy implications

Our results suggest strengthening the partnership between mobile telephony and banking institutions for the provision of mobile banking services. Financial institutions can rely on mobile phone channels to offer financial services, such as micro-savings products, adapted to the modest working and income conditions of most economic agents. To improve savings mobilization, service providers must eliminate exclusionary savings conditions, such as minimum balances, registration fees, and transfer fees. And to improve the adoption of mobile banking services, it is necessary to increase users' education, create jobs, and guarantee access to a decent income.

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## Author statement

The author declare to have written this manuscript in full and declare that this manuscript is not submitted to any other journal.

## Declaration of Competing Interest

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